Convex Subdivision

A project for
Computer Geometry
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Abstract

This project aims to take on the problem of subdividing a concave polygon into as few convex polygons as possible. The idea is to triangulate the polygon first, as triangles are always convex. Afterwards, remove edges to form bigger convex polygons. To do this, a variant of a well-known algorithm, the ear-clipping algorithm, was implemented. Compared to simply triangulating the entire polygon, the algorithm reduced the number of edges used, and thus the number of sub-polygons made by about 40%. However, this project's algorithm is still imperfect, and there are further ways to improve it.

The method mentioned above, a variant on an existing algorithm, runs by continuing to clip ears next to already clipped ears and recording them as a single polygon, instead of randomly selecting an ear. This simple variation combines both triangulation and edge reduction into a single step, increasing efficiency.

As for improvements, the data structure itself is basic, and a major limiting factor is the fact that this algorithm is based on a simple algorithm. Sometimes, the algorithm will leave behind edges that can be removed. A half-edge data structure may increase complexity, but in turn allow for more accurate subdivisions.

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Problem Definition

The problem is simple. Take a polygon, and subdivide it with as few edges as possible, so that every sub-polygon is convex.

In computer applications, a convex polygon is easier to process than a concave one. Thus, for simplicity, a concave polygon can be thought of as a collection of convex polygons, and each one is then processed individually. This idea can be seen in many places, such as in 3D models, where every object is a collection of triangles, which are always convex. It can also be seen in games, such as fighters, where a hitbox or hurtbox is not a single polygon, but multiple rectangles or circles in 2D, which are convex. For 3D, the equivalent are boxes and spheres, which are also convex.

To put simply, this problem is about creating a general algorithm to turn a concave polygon into a convex one to make processing easier.

Input	Output
A polygon's vertices, ordered counter-clockwise.	A list of edges, zero-indexed, each indicating the start and end vertices.
The polygon must be simple. It must not cross itself at any point.	The output may not be perfect.